Sanskar Rajesh Mankar

Quantitative Researcher

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Portfolio: sanskarmankar.github.io/Portfolio

SUMMARY

Quantitative researcher building systematic trading strategies for Indian equity indices and crypto. Developed BTST and intraday strategies with backtested Sharpe ratios of 2.9–3.2 on NIFTY 50, and a Hawkes-process BTC alpha that ranked among the best-performing strategies on Untrade. Focused on testing whether an edge is repeatable and robust, using out-of-sample, walk-forward and Monte Carlo validation. Background in end-to-end ML engineering (LLM fine-tuning, RAG, computer vision).

TECHNICAL SKILLS

Quantitative Research: Alpha Research, Signal Generation, Hawkes Process, Volatility Clustering, Time-Series Analysis, Statistical Modeling, Genetic Algorithms, Monte Carlo Simulation

Validation & Backtesting: Event-Driven Backtesting, Out-of-Sample Testing, Walk-Forward Testing, Look-Ahead Bias Prevention, Parameter-Stability Analysis, Risk Analysis

Programming & Data: Python, Pine Script, NumPy, Pandas, ThreadPoolExecutor (Parallel Processing), OHLCV Data Pipelines

Machine Learning: PyTorch, FAISS, YOLOv8, CUDA, FastAPI

PROFESSIONAL EXPERIENCE

Independent Quantitative Researcher | Self-Employed, Remote **07/2025 – Present**

- Research and backtest systematic strategies across Indian equity indices and equities, validating whether each edge is repeatable across instruments and market regimes.
- Built in-house research infrastructure: event-driven backtesting, look-ahead bias prevention, out-of-sample and walk-forward testing, Monte Carlo robustness analysis and parameter-stability testing.
- Screened 300+ equities using scalable OHLCV processing pipelines; identified that the BTST signal does not hold on cash equities and scoped it to index instruments. Current research: stock swing-trading systems and BTC options strategies.

Quantitative Researcher | Untrade, Jaipur, India **02/2024 – 07/2025**

- Developed a BTC alpha using Hawkes process modeling to detect volatility expansion and predict direction: **68% win rate, 1:3 risk-to-reward, Sharpe 6+** (backtest). Ran live and ranked among the best-performing strategies on the platform.
- Built Ethereum alpha models using event-driven signals and systematic parameter optimization.
- Engineered a 5-island genetic algorithm with adaptive mutation and parallel evaluation (ThreadPoolExecutor) for faster parameter exploration.
- Created research protocols for look-ahead bias detection, strict in-sample/out-of-sample validation and Monte Carlo robustness testing.
- Automated the research pipeline: data extraction, signal generation, backtesting, validation and risk analysis.

AI/ML Engineer Intern | Rajyug Solutions, Pune, India **09/2023 – 01/2024**

- Fine-tuned an 8B-parameter model for medical Q&A and served it with CUDA-accelerated inference; deployed ML applications as FastAPI services.
- Built a FAISS-based RAG pipeline to ground responses in source PDFs.
- Developed a YOLOv8 helmet detection system with **97% accuracy**; built emotion classification and resume-screening models.

SELECTED RESEARCH PROJECTS

BTST / Overnight Long-Only Strategy | NIFTY 50, Midcap Select, Next 50 **Python, Pine Script**

- **NIFTY 50 backtest:** 946 trades, 59.8% win rate, 22.95% CAGR, Sharpe 3.21, Sortino 6.20, Profit Factor 1.90, max drawdown -3.81%.
- Approximately 25% CAGR on NIFTY Midcap Select and 30%+ on NIFTY Next 50. Signal adaptable to index futures and directional option strategies.

Intraday Long/Short Directional Strategy | NIFTY 50, Midcap Select, Next 50 **Python**

- **NIFTY 50 backtest:** 2,365 trades, 47.1% win rate, 31.18% CAGR, Sharpe 2.86, Sortino 5.98, Profit Factor 1.60, max drawdown -6.10%, 1.80 payoff ratio, ~134 min average hold.
- Profitable on both long and short sides, with each contributing independently to returns.

All performance figures are from historical backtests, not live trading.

EDUCATION

B.Tech, Information Technology | MIT ADT University, Pune **2023**